

Alex Batlle Casellas

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EXPERIENCE

Machine Learning Engineer	Qualcomm Europe	Remote	Sep 2023 - Present
◦ Research on ML infrastructure and LLM reasoning. Led work on multi-node LLM pretraining on clusters ranging from 16×8xA100 to 60×8xH100 nodes, focusing on RDMA/RoCE-based communication, NCCL collective tuning, and performance portability, demonstrating near-linear scalability on Ethernet (RoCE) clusters, rivalling InfiniBand at scale. Results in two peer-reviewed publications at SC Workshops '25, one of which received the Best Paper Award (INDIS). Workloads managed via Run:ai (Kubernetes-native GPU scheduling). Additional work on efficient training techniques including quantization-aware training and knowledge distillation at HPC scale (P6), wireless network modeling (P1), and LLM reasoning (P4, P5).			
Visiting Student	TU Delft	Delft, South Holland, NL	Feb 2023 - Sep 2023
◦ Research visit to write my Bachelor's degree Final Thesis on autonomous robot exploration via Graph Neural Networks and Reinforcement Learning, specifically on learning exploration policies on graph-structured map representations. Available online .			
Software Engineering Intern	Qualcomm Europe	Remote	Jun 2022 - Feb 2023
◦ Worked on a combinatorial optimization problem for telecommunications equipment placement in urban networks. ◦ Developed a method for a network digital twin using GNNs.			
Software Engineering Intern (Robotics)	HP	Remote	Summer 2021
◦ Built a graphical application for ROS job visualization as a directed graph prior to deployment, using C++ and Qt.			

PUBLICATIONS

- High-Performance Modeling and Optimization of Wireless Networks Using Bottleneck Structures (2025)** [doi:10.1109/ACCESS.2025.3586752](https://doi.org/10.1109/ACCESS.2025.3586752)
Aleix Torres-Camps, [Àlex Batlle Casellas](#), Yuyang Wang, Mauro Filomeno Rivero, Naga Bhushan, Jordi Ros-Giralt
IEEE Access (Vol 13)
- Scaling LLM Training Using RDMA over Converged Ethernet (2025)**, [doi:10.1145/3731599.3767453](https://doi.org/10.1145/3731599.3767453)
[Àlex Batlle Casellas](#), Adrián Pérez Diéguez, Aleix Torres-Camps, Harris Teague, Arnau Padrés Masdemont, Jordi Ros-Giralt
SC Workshops '25: INDIS \  *Best paper award*
- Pretraining LLMs at Scale: Tuning Strategies and Performance Portability (2025)**, [doi:10.1145/3731599.3767699](https://doi.org/10.1145/3731599.3767699)
Adrián Pérez Diéguez, [Àlex Batlle Casellas](#), Aleix Torres-Camps, Harris Teague, Jordi Ros-Giralt
SC Workshops '25: PMBS
- M3Kang: Evaluating Multilingual Multimodal Mathematical Reasoning in Vision-Language Models (2026)** [arxiv:2601.16218](https://arxiv.org/abs/2601.16218)
Aleix Torres-Camps, Nathaniel Mitrani Hadida, Víctor Conchello Vendrell, [Àlex Batlle Casellas](#), Arnau Padrés Masdemont, Jordi Ros-Giralt
arXiv preprint
- LaneRoPE: Positional Encoding for Collaborative Parallel Reasoning and Generation (ICLR 2026 MALGAI)** [OpenReview Forum](#)
Gabriele Cesa, Thomas Hehn, Aleix Torres-Camps, [Àlex Batlle Casellas](#), Jordi Ros-Giralt, Arash Behboodi, Tribhuvanesh Orekondy
- Optimizing Teacher-Student Partitioning for Scalable Knowledge Distillation on HPC Systems (2026)** (pending publication at ISC2026 - AI on HPC)

EDUCATION

M.S. in Mathematics	Sep 2024 - Jun 2026
FME - Universitat Politècnica de Catalunya	Barcelona, CAT
Interested in: Graph Theory, Computational Complexity, Algorithmic Game Theory, Combinatorics, HPC4AI	
Double B.S. in Mathematics + Data Science and Engineering	Sep 2018 - Oct 2023
CFIS - Universitat Politècnica de Catalunya	Barcelona, CAT
Majoring in Mathematics and Data Science and Engineering in a double degree, 5-year study plan within CFIS, a highly selective excellence center within UPC (~35 students admitted per year across several STEM disciplines).	

SKILLS

Programming Languages:  Python ( PyTorch,  NumPy,  Transformers,  Datasets,  Accelerate),  C++,  Bash.

ML Infrastructure: Multi-node distributed training, RDMA/RoCE/InfiniBand, NCCL collective optimization, GPU-CPU communication profiling, Kubernetes GPU scheduling (Run:ai Scheduler), HPC cluster scheduling (Slurm, via BSC MareNostrum access), performance benchmarking.

Tools:  Git,  Linux,  Claude Code,  OpenClaw (skill authoring).

Languages: Catalan (native), English (proficient), Spanish (bilingual)